



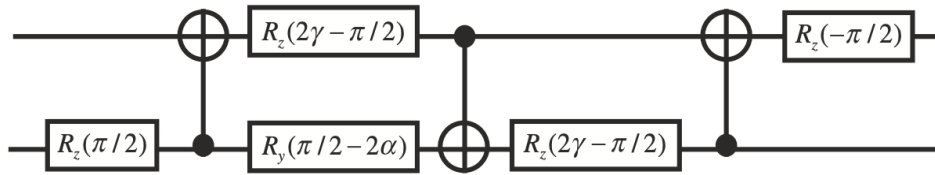
PEP 559

Machine Learning in Quantum Physics

Dr. Chunlei Qu

Spring 2026

Reinforcement learning for quantum computing (quantum control optimization)

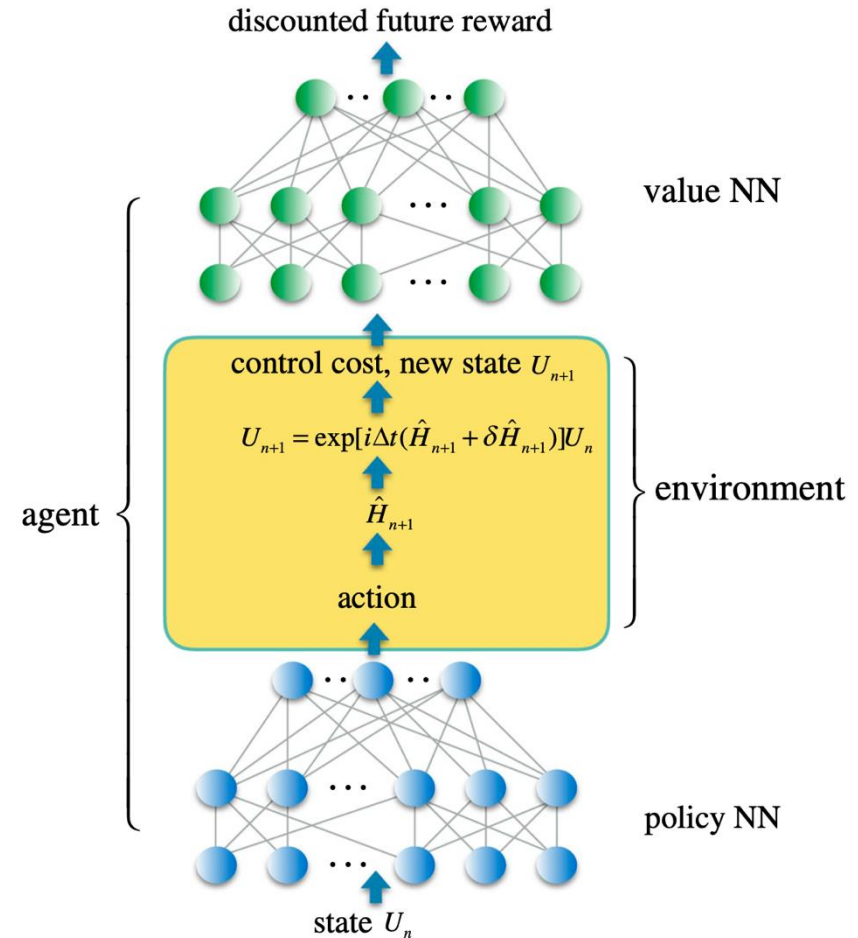


A quantum circuit

- Depth: 7
- Three two-qubit gates
- Five single qubit gates

To realize a unitary transformation

$$\mathcal{N}(a, \alpha, \gamma) = \exp[i(\alpha\sigma_1^x\sigma_2^x + \alpha\sigma_1^y\sigma_2^y + \gamma\sigma_1^z\sigma_2^z)],$$



How do we find the optimal parameters to realize such an operation with the above circuit?

Outline

- I. Quantum computing (qubits, gates, circuits, etc)
- II. Introduction to reinforcement learning
- III. RL for quantum gate optimization



Amplitudes and Probabilities

$$\Psi(\mathbf{x}, t)$$

$$|\Psi(\mathbf{x}, t)|^2$$

Born's Rule:

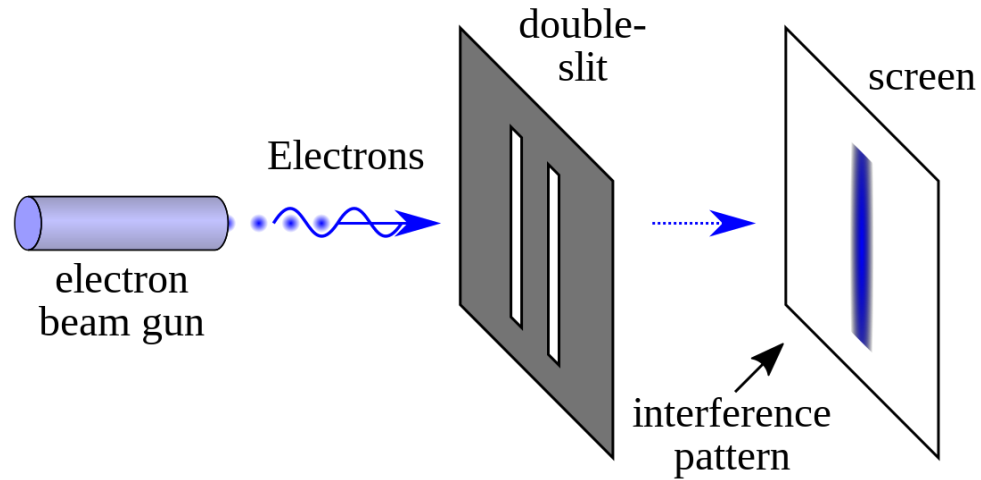
- Quantum systems are described by complex wave functions (the probability amplitudes)
- The probabilities of finding a particle at a point $\mathbf{x}$ is the absolute square of the wave function



Quantum Interference

The double-slit experiment

Since there are two paths, we first add the amplitudes, and then take the absolute square of the sum to find the probability



$$\begin{aligned} p &= |z|^2 \\ &= |z_1 + z_2|^2 \\ &= |z_1|^2 + |z_2|^2 + z_1^* z_2 + z_1 z_2^* \\ &= p_1 + p_2 + |z_1||z_2| \left(e^{i(\varphi_2 - \varphi_1)} + e^{-i(\varphi_2 - \varphi_1)} \right) \\ &= p_1 + p_2 + \underbrace{2\sqrt{p_1 p_2} \cos(\varphi_2 - \varphi_1)}_{\text{interference terms}} \end{aligned}$$

where

$$z_1 = |z_1| e^{i\varphi_1}$$

$$z_2 = |z_2| e^{i\varphi_2}$$

Originates from
the addition of
wave functions

Quantum interference

$$p = p_1 + p_2 + \underbrace{2\sqrt{p_1 p_2} \cos(\varphi_2 - \varphi_1)}_{\text{interference terms}}$$

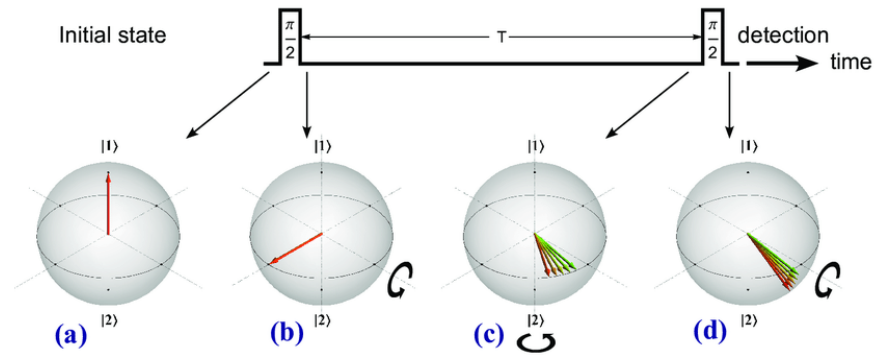
Negative interference term $\rightarrow$ **destructive** interference

Positive interference term $\rightarrow$ **constructive** interference

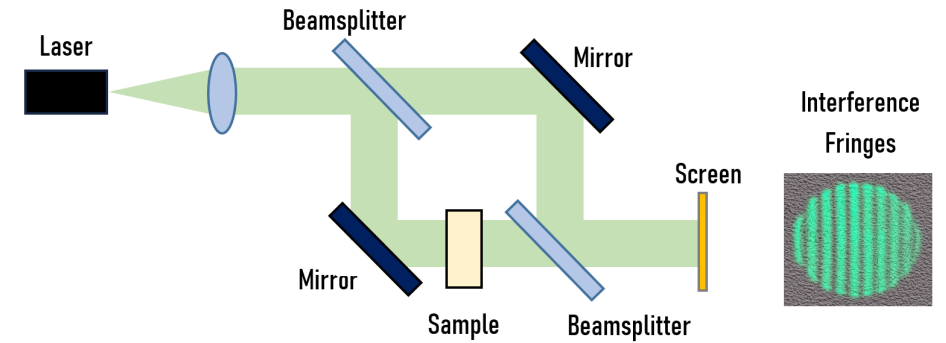
Even for single quantum particle, we cannot say it has travelled through either the upper or the lower slit. It has travelled through both.

Quantum superposition (in terms of Dirac notations)

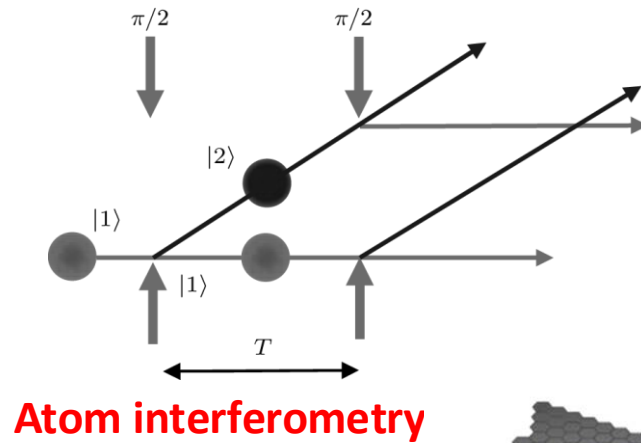
$$|\psi\rangle = \alpha|\text{upper slit}\rangle + \beta|\text{lower slit}\rangle$$



Ramsey interferometry

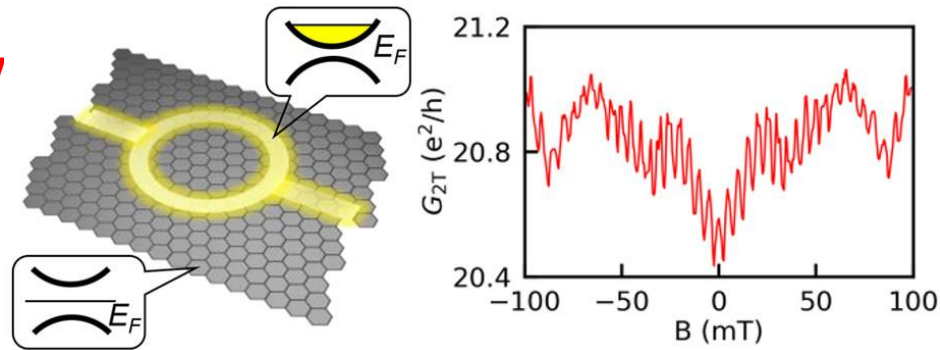


Mach-Zehnder interferometry

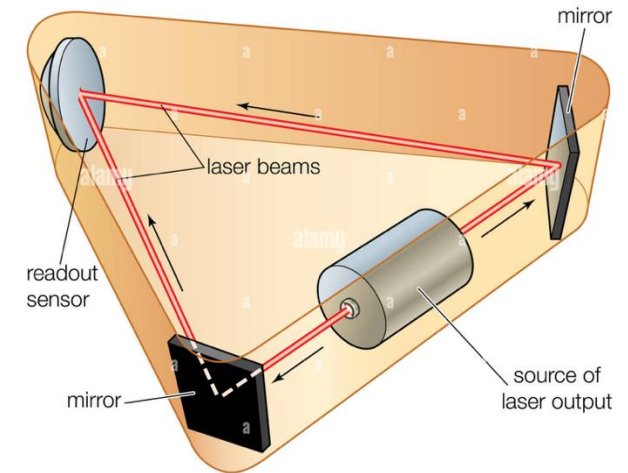


Atom interferometry

Quantum interferometer



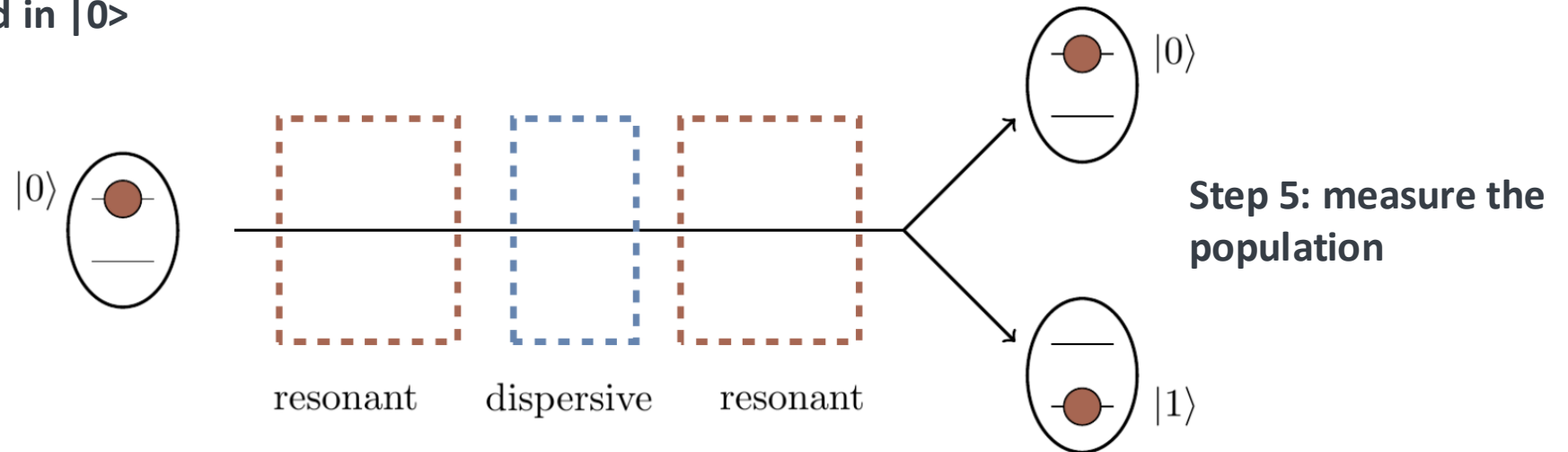
Aharonov-Bohm interferometry



Ring laser gyroscope

Ramsey interferometry

Step 1: Two-level atoms, initially prepared in $|0\rangle$

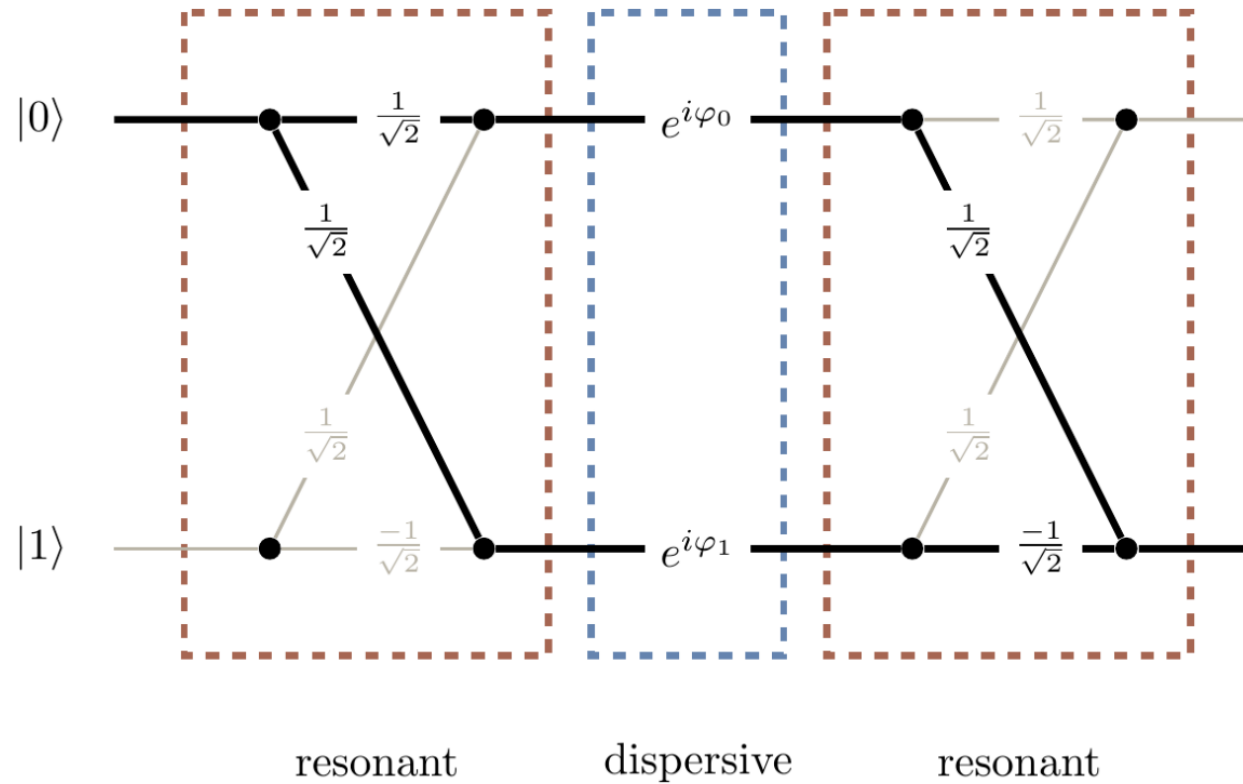


Step 2: apply a resonant $\pi/2$ -pulse of Rabi-coupling

Step 3: the atom acquires a phase shift (different phase accumulation for the two levels)

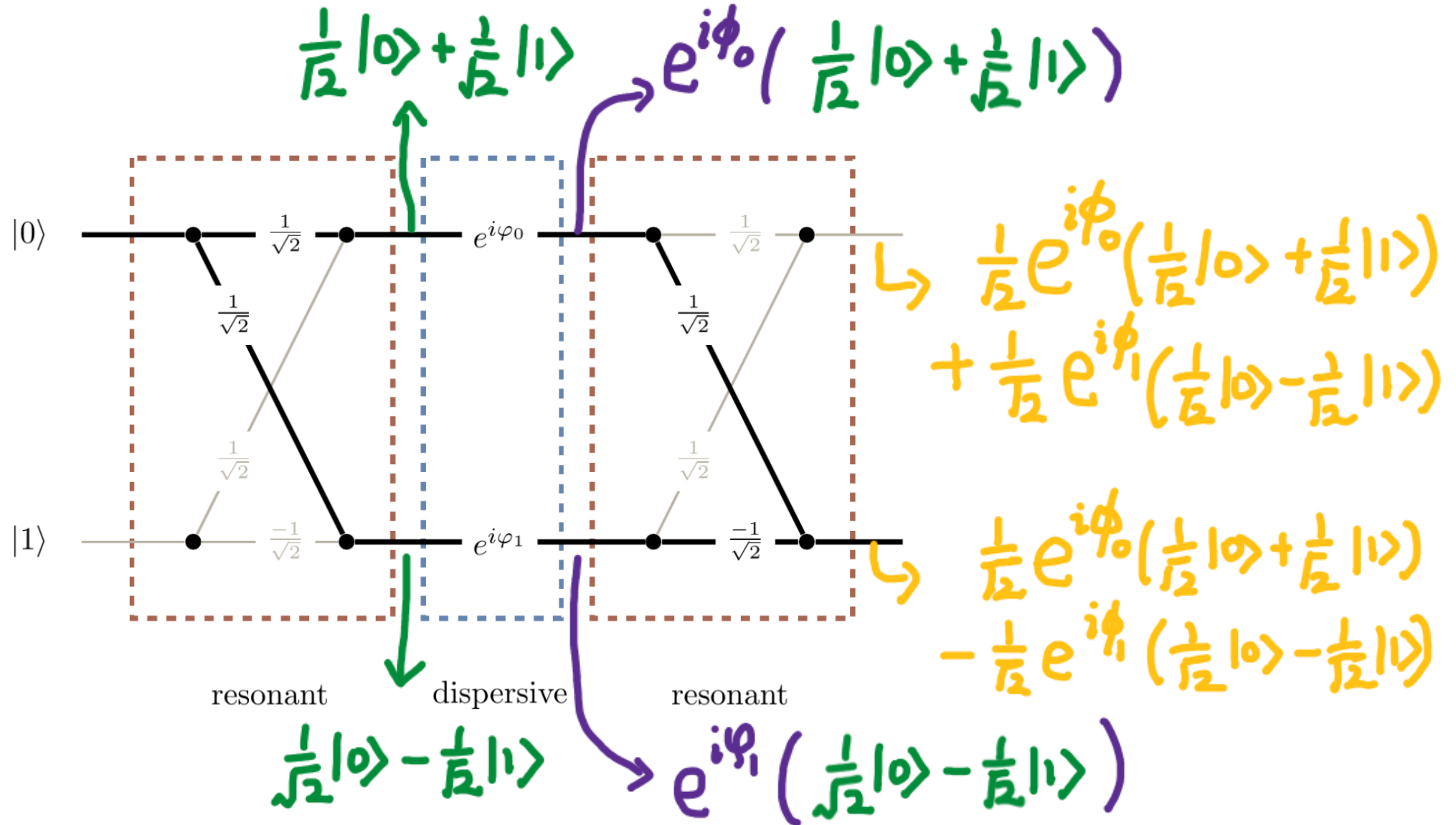
Step 4: apply a resonant $\pi/2$ -pulse of Rabi-coupling

Ramsey interferometry



Similar to the two-slit experiment, there are two possible paths from left to right

Ramsey interferometry



Initially in $|0\rangle$

The amplitude to find the atom in $|1\rangle$ state will be

$$\begin{aligned} U_{10} &= \frac{1}{\sqrt{2}} e^{i\varphi_0} \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} e^{i\varphi_1} \frac{-1}{\sqrt{2}} \\ &= \frac{1}{2} (e^{i\varphi_0} - e^{i\varphi_1}). \end{aligned}$$

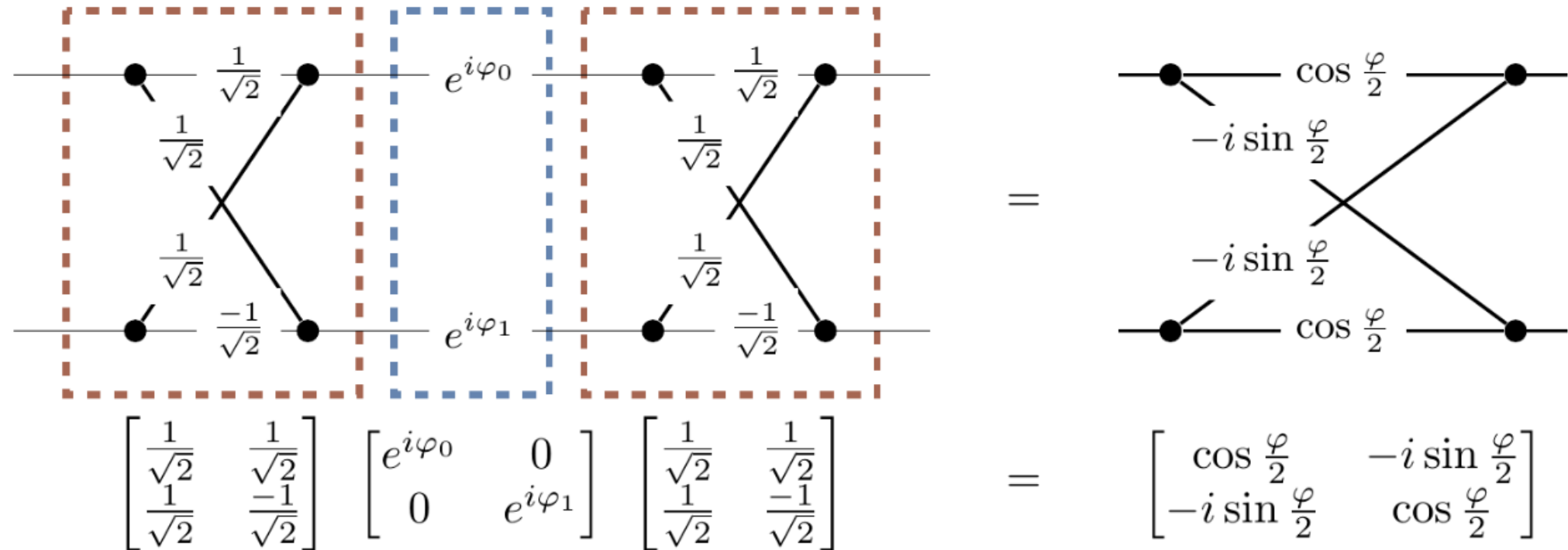
The corresponding probability will be

$$\begin{aligned} P_{10} &= |U_{10}|^2 \\ &= \left| -ie^{i\frac{\varphi_0+\varphi_1}{2}} \sin \frac{\varphi_1 - \varphi_0}{2} \right|^2 \\ &= \left| \sin \frac{\varphi_1 - \varphi_0}{2} \right|^2 \\ &= \frac{1}{2} (1 - \cos \varphi) \end{aligned}$$

We can verify that the probability of finding the particle in $|0\rangle$ is

$$P_{00} = 1 - P_{10}$$

Matrix representation



Each part is like a device (a quantum operator that is represented as a matrix)
It acts on the initial state to get the final state

Quantum bits (qubits)

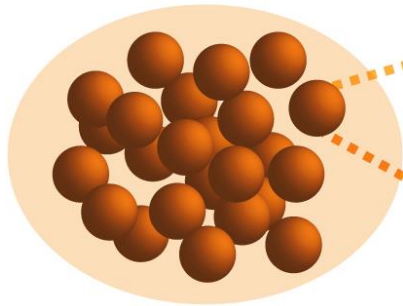
- Two well defined quantum states
- They can be coupled together, and they are isolated from other states
- Such two-level quantum systems are called quantum bits, or qubits



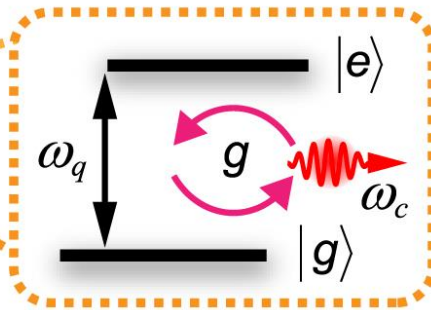
Examples of qubits

Internal states of atoms

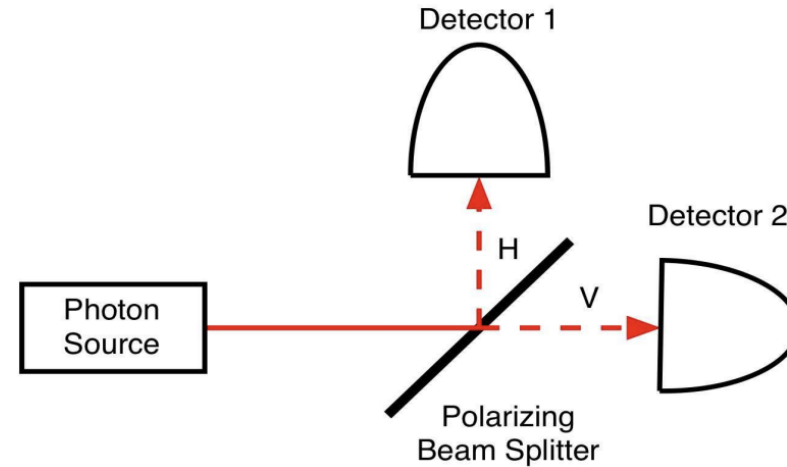
atomic ensemble



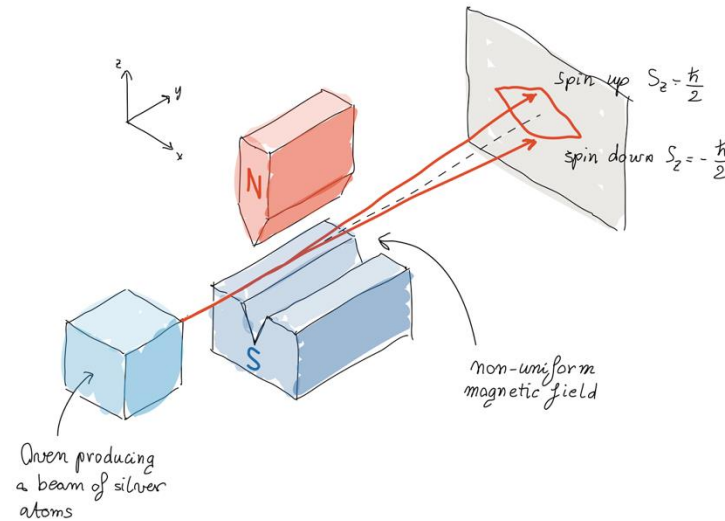
atomic level



Polarization of photons

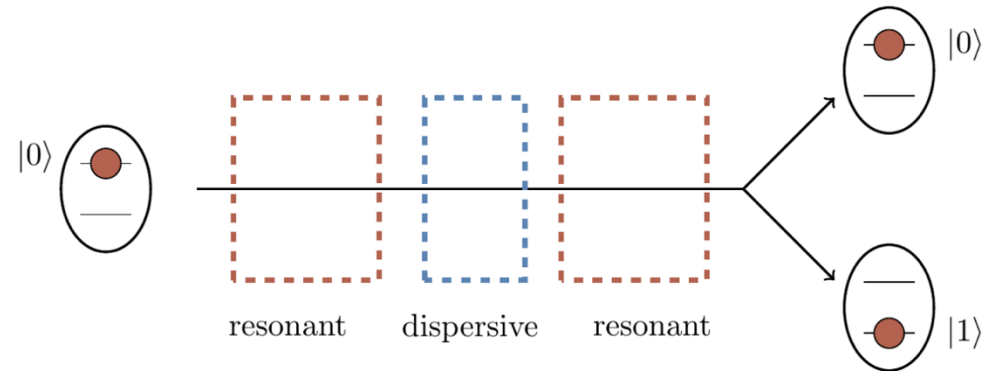


Spin states of electrons



Quantum gates

Ramsey interferometry is to apply a series of quantum transformations or operations (quantum gates) on the qubit



$$|0\rangle \xrightarrow{H} \bullet^\varphi \xrightarrow{H} \cos \frac{\varphi}{2} |0\rangle - i \sin \frac{\varphi}{2} |1\rangle$$

Initial state $|0\rangle$

Amplitude in $|0\rangle$ at the end of the circuit

Hadamard gates

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

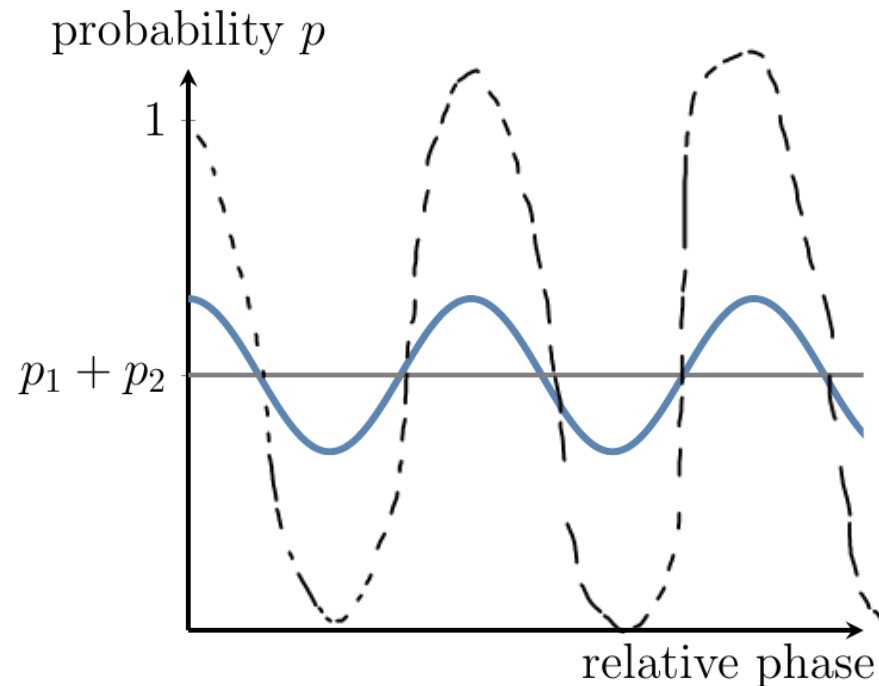
Phase gate

$$P_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}$$

Quantum decoherence

We can hardly see quantum interference in the daily life. Why?

Perfect quantum oscillation only occurs in isolated quantum systems. **Decoherence suppresses quantum interference.**



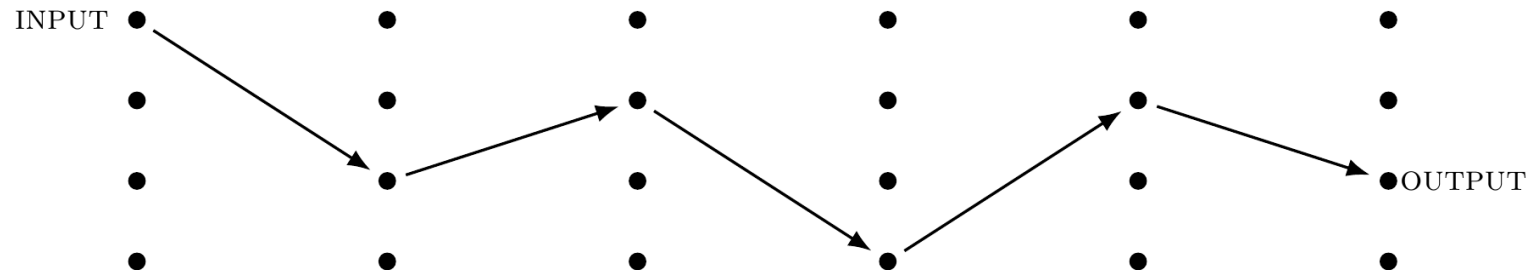
$$p = p_1 + p_2 + 2v\sqrt{p_1 p_2} \cos(\varphi_2 - \varphi_1)$$

v: visibility of the interference pattern

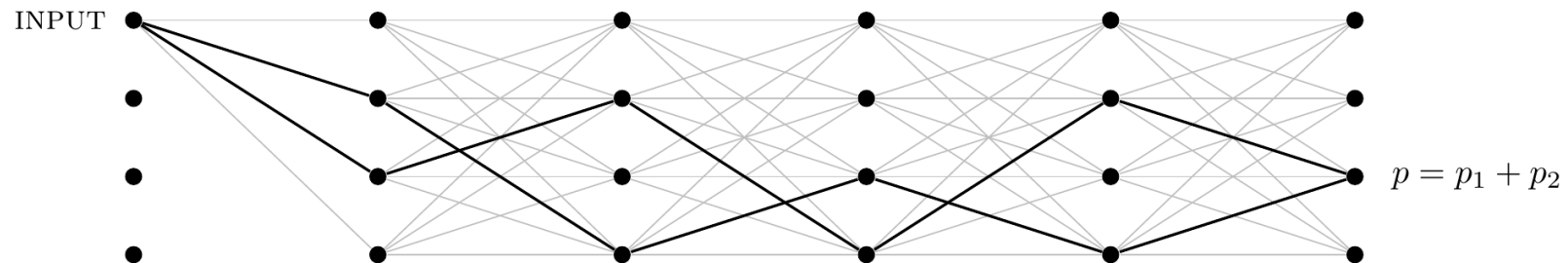
- v=0: total decoherence
- v=1: no decoherence (full interference)
- 0<v<1: partial decoherence

Computations

Deterministic computation

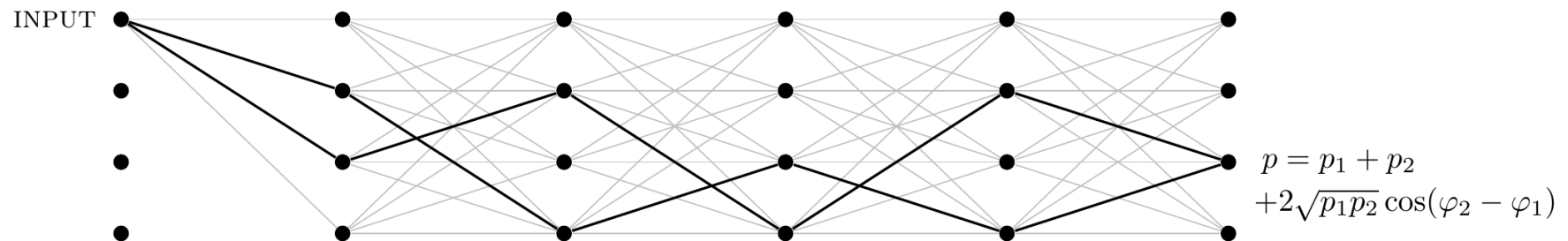


Probabilistic computation



e.g.,
Monte-Carlo

Quantum computation

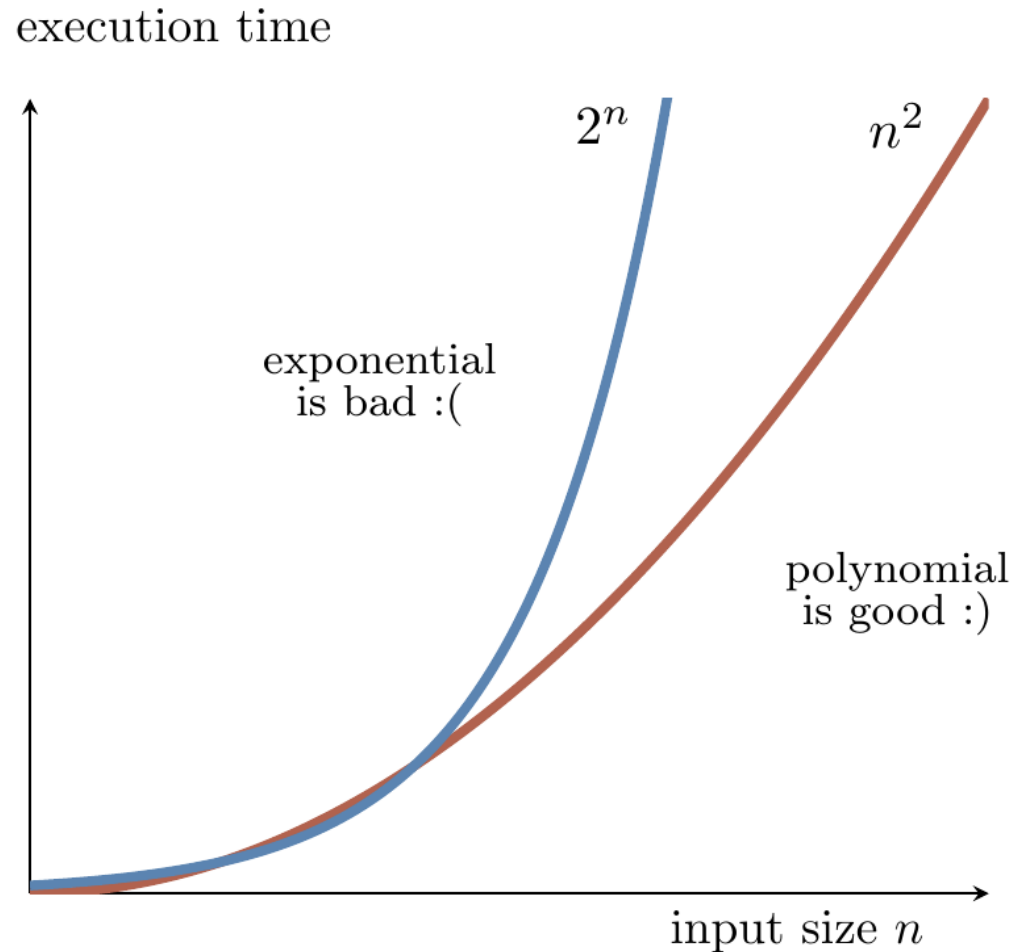


Quantum computation

Quantum computation can be viewed as a **complex multi-particle quantum interference** involving many computational paths through a computing device.

The art of quantum computation is to **shape the quantum interference through a sequence of computational steps**, enhancing probabilities of the “correct” outputs and suppressing probabilities of the “wrong” ones.

Computational complexity



In order to solve a particular problem, computers (classical or quantum) follow a precise set of instructions called an **algorithm**

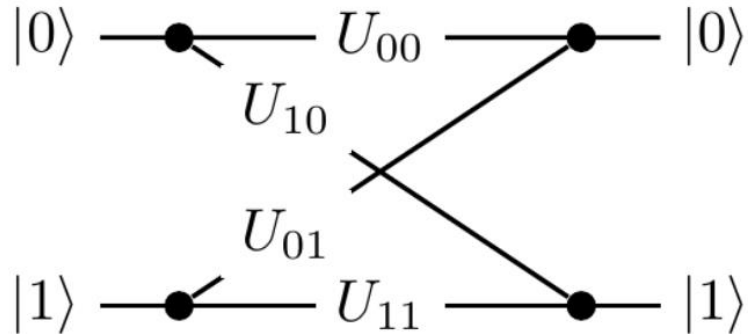
An algorithm is said to be **efficient** if the number of elementary operations taken to execute it increases no faster than **polynomial function of the size of the input**

Number factoring with classical computer is **exponential**

Peter Shor showed that quantum computer can solve it with a computational time $\sim n^2$

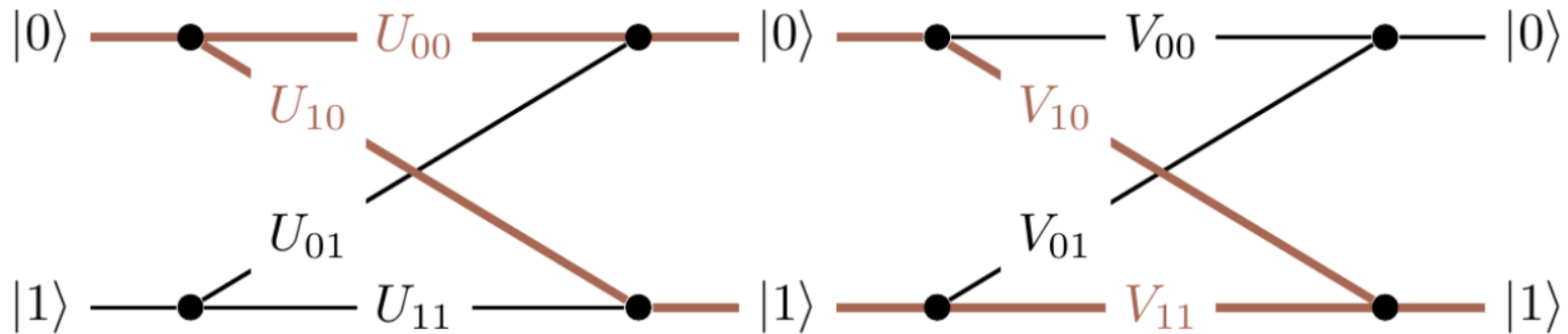
Unitary operations

Each computational step is to act a 2x2 unitary matrix on the qubit



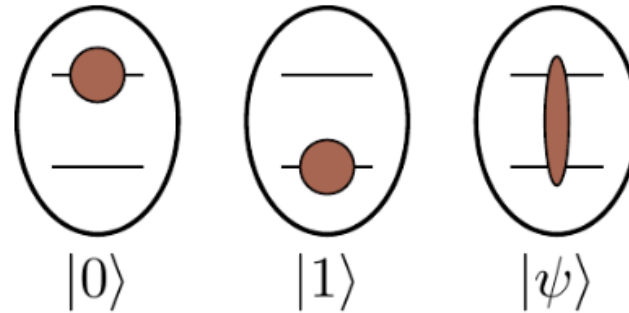
$$U = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix}$$

The composition of two computational step



Abstract notations for Qubits and Quantum Gates

$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle \leftrightarrow \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix}$$



$$|\psi\rangle \mapsto |\psi'\rangle = U|\psi\rangle$$

$$\begin{bmatrix} \alpha'_0 \\ \alpha'_1 \end{bmatrix} = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix}$$

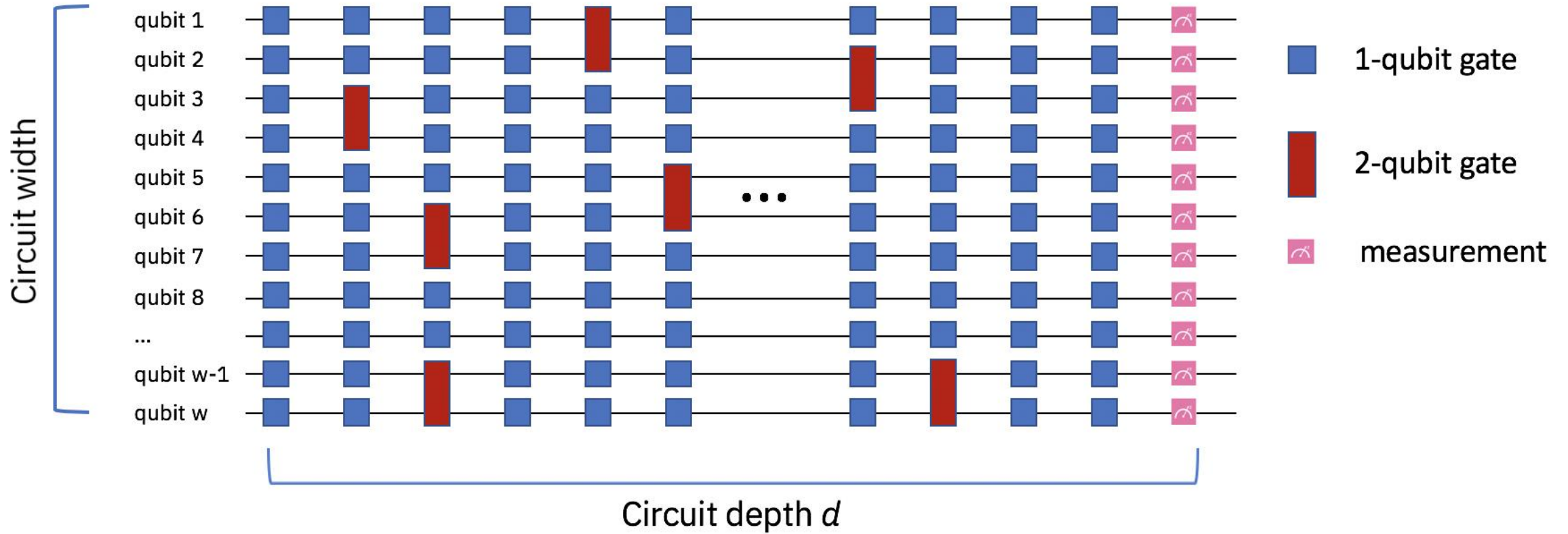
Quantum circuits

- A **quantum circuit** is a device consisting of a series of quantum logic gates whose computational steps are synchronized in time
- The **size of a circuit** is the number of gates it contains
- The gates in a circuit can be divided into **layers**, where the gates in the same layer operate at the same time
- The number of such layers is called the **depth of a circuit**



Quantum circuits

A quantum circuit of width w and depth d

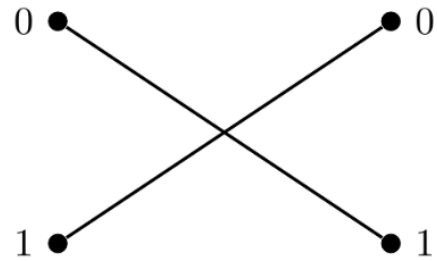


Some important quantum gates



NOT and The square root of NOT

NOT gate



NOT

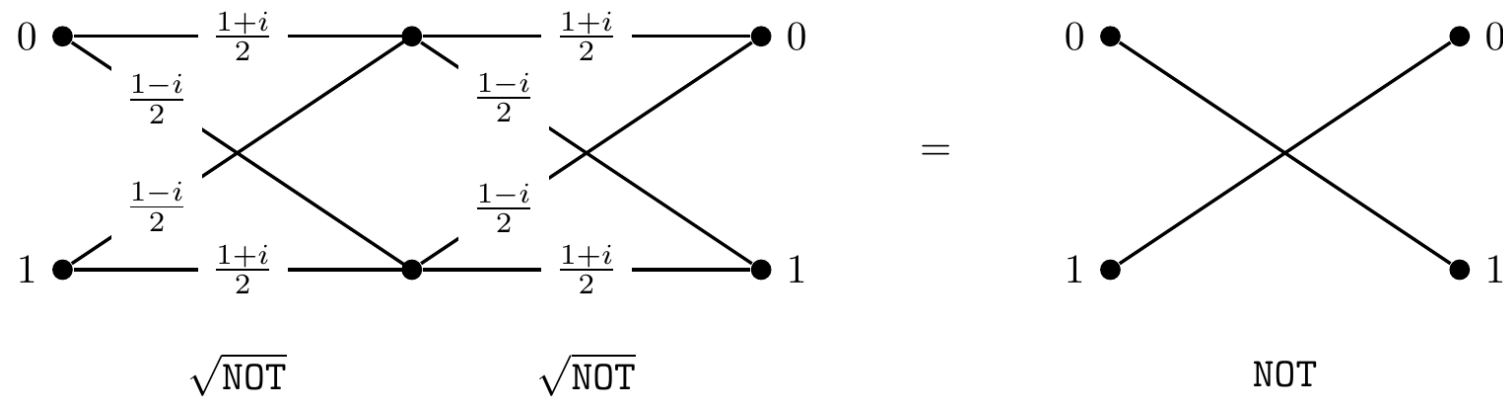
$$\text{NOT} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sqrt{\text{NOT}} = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} e^{i\frac{\pi}{4}} & e^{-i\frac{\pi}{4}} \\ e^{-i\frac{\pi}{4}} & e^{i\frac{\pi}{4}} \end{bmatrix}$$

The square root of NOT

Check

$$\frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$|0\rangle \xrightarrow{\sqrt{\text{NOT}}} \frac{1}{\sqrt{2}} \left[e^{i\frac{\pi}{4}} |0\rangle + e^{-i\frac{\pi}{4}} |1\rangle \right] \xrightarrow{\sqrt{\text{NOT}}} |1\rangle$$

Phase gates

$$P_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix} \quad \text{or} \quad P_\varphi = \begin{bmatrix} e^{-i\frac{\varphi}{2}} & 0 \\ 0 & e^{i\frac{\varphi}{2}} \end{bmatrix} \quad \text{Global phase not important}$$

More specifically, we may use the following phase gates

Phase-flip	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{array}{l} 0\rangle \mapsto 0\rangle \\ 1\rangle \mapsto - 1\rangle \end{array}$	Sometimes it is also called Pauli-Z gate
$\frac{\pi}{4}$ -phase	$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	$\begin{array}{l} 0\rangle \mapsto 0\rangle \\ 1\rangle \mapsto i 1\rangle \end{array}$	
$\frac{\pi}{8}$ -phase	$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$	$\begin{array}{l} 0\rangle \mapsto 0\rangle \\ 1\rangle \mapsto e^{i\frac{\pi}{4}} 1\rangle \end{array}$	

Pauli operators/gates

σ_x, σ_y , and σ_z , also denoted by X, Y , and Z ,

They do not commute

$$XY = -YX,$$

$$XZ = -ZX,$$

$$YZ = -ZY.$$

They are related

$$ZX = iY$$

NOT
gate

Identity	$\mathbf{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$ 0\rangle \mapsto 0\rangle$ $ 1\rangle \mapsto 1\rangle$
Bit-flip	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$ 0\rangle \mapsto 1\rangle$ $ 1\rangle \mapsto 0\rangle$
Bit-phase-flip	$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	$ 0\rangle \mapsto i 1\rangle$ $ 1\rangle \mapsto -i 0\rangle$
Phase-flip	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$ 0\rangle \mapsto 0\rangle$ $ 1\rangle \mapsto - 1\rangle$



From bit-flits to phase-flips, and back again

by applying the Hadamard gate, we can change Z (X) gate to X (Z) gate

$$\text{---} \boxed{H} \text{---} \boxed{Z} \text{---} \boxed{H} \text{---} = \text{---} \boxed{X} \text{---}$$

They are useful for quantum error correction

$$\text{---} \boxed{H} \text{---} \boxed{X} \text{---} \boxed{H} \text{---} = \text{---} \boxed{Z} \text{---}$$

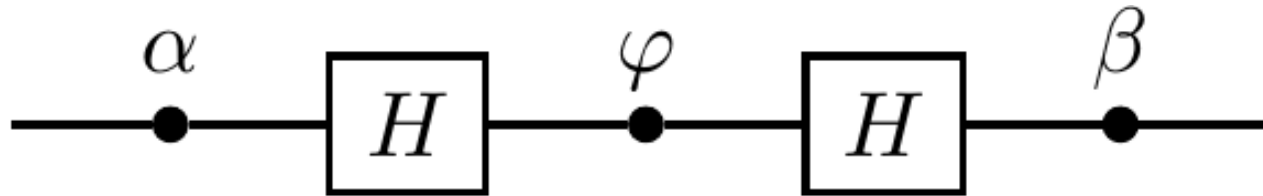
$$H X H = Z$$

$$H Z H = X$$

$$H Y H = -Y$$

Single-qubit quantum gates

In principle, we may have **infinitely number** of single-qubit quantum unitary operations



Any unitary operation on a qubit (up to an overall phase) can be implemented by the above circuit, containing **two Hadamard gates and three phase gates** (with adjustable phase settings)

$$U(\alpha, \beta, \varphi) = \begin{bmatrix} e^{-i(\frac{\alpha+\beta}{2})} \cos \varphi/2 & -ie^{i(\frac{\alpha-\beta}{2})} \sin \varphi/2 \\ -ie^{-i(\frac{\alpha-\beta}{2})} \sin \varphi/2 & e^{i(\frac{\alpha+\beta}{2})} \cos \varphi/2 \end{bmatrix}$$

Quantum dynamics

Schrödinger equation:

$$\frac{d}{dt}|\psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\psi(t)\rangle.$$

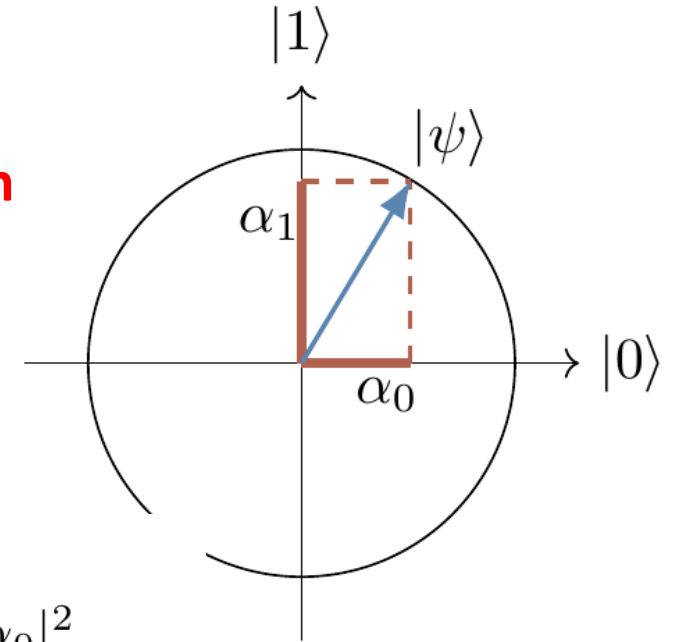
For time-independent Hamiltonian

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\psi(0)\rangle.$$

Quantum measurement

Quantum measurement is **not** a **passive acquisition of information**

If we **measure** a quantum system, we **disturb** it.



$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle \longrightarrow \boxed{\text{Measurement}} \longrightarrow \begin{cases} |0\rangle & \text{with probability } |\alpha_0|^2 \\ |1\rangle & \text{with probability } |\alpha_1|^2 \end{cases}$$

The sudden change of quantum state, from the pre-measurement state to the post-measurement state, is called a **collapse**

Quantum entanglement

From one qubit to two

Single qubit basis

$$|0\rangle, |1\rangle$$

The most general state of one qubit $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$

Two qubits basis

$$\begin{aligned} |00\rangle &\equiv |0\rangle|0\rangle & |01\rangle &\equiv |0\rangle|1\rangle \\ |10\rangle &\equiv |1\rangle|0\rangle & |11\rangle &\equiv |1\rangle|1\rangle \end{aligned}$$

The most general state of two qubits (a bipartite state)

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$$

Examples of 2-qubit states

Is there any essential difference between them?

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle$$

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Separable vs. Entangled states

Separable if it can be decomposed into a pair of state vectors where each one pertains to one of the two qubits only

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle = \frac{1}{\sqrt{2}} \underbrace{|0\rangle}_{\text{qubit 1}} \underbrace{(|0\rangle + |1\rangle)}_{\text{qubit 2}}$$

Entangled if it does not admit such a decomposition

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \not= |\psi_1\rangle|\psi_2\rangle$$

More qubits

Separable 3-qubit states

$$|0\rangle \otimes |1\rangle \otimes |1\rangle \equiv |011\rangle$$

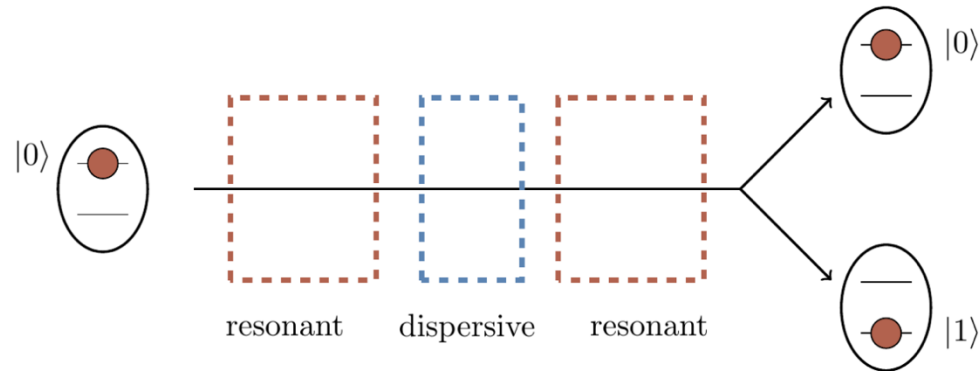
$$|1\rangle \otimes |1\rangle \otimes |1\rangle \equiv |111\rangle$$

Apply the Hadamard gate to the first qubit, it is still separable

$$\begin{aligned} |011\rangle &\xrightarrow{H \otimes 1 \otimes 1} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |1\rangle \otimes |1\rangle \\ &= \frac{1}{\sqrt{2}} (|011\rangle + |111\rangle). \end{aligned}$$

Hadamard gate $\rightarrow$ quantum interference

For single qubit, it does not create entanglement between qubits



$$|0\rangle \xrightarrow{H} \bullet^\varphi \xrightarrow{H} \cos \frac{\varphi}{2} |0\rangle - i \sin \frac{\varphi}{2} |1\rangle$$

Initial state $|0\rangle$

Amplitude in $|0\rangle$ at the end of the circuit

Hadamard gates

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

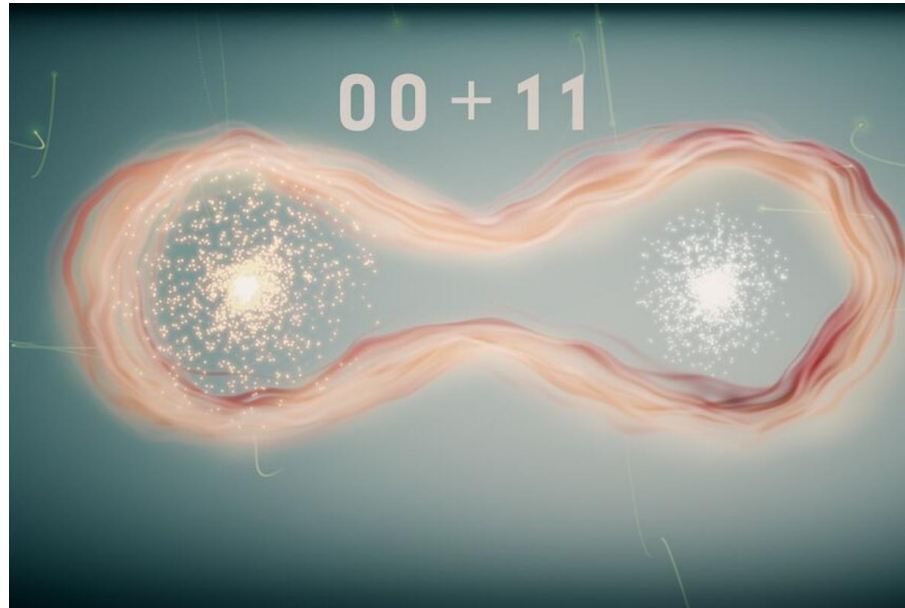
Phase gate

$$P_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}$$

How to create entangled states?

Entanglement is the result of interactions

Applying single-qubit gates to change individual qubits won't create entanglement



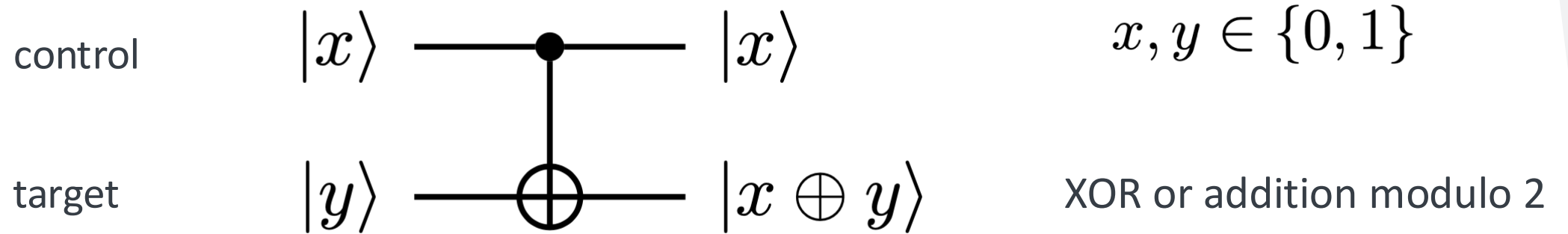
controlled-NOT (c-NOT)

- The most popular two-qubit entangling gate
- Also known as controlled-X gate
- It acts on two qubits: it flips the 2nd qubit (target) if the first qubit (control) is $|1\rangle$, and does nothing if the control qubit is $|0\rangle$

In the standard basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$$\left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

C-NOT gate circuit notation



input		output	
x	y	x	y+x
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$